



Pediatric Pearls:

- ### Signs & Symptoms:

- Differential:**

- Hypovolemia
- Sepsis
- Neurogenic Shock
- Anaphylaxis
- Dysrhythmia
- Device malfunction / Failure

- Contact patient's LVAD coordinator as early as possible
- Perform general [patient assessment](#)
- [Oxygen](#) PRN titrated to SpO₂ 92% - 96%
- Basic airway management
- Ensure LVAD has power supply
- If LVAD failure is noted and no signs of life, begin [CPR procedure](#). Refer to appropriate COG – Defibrillator can be used as normal, shift pads to not place directly on device.

- 4 lead and [12 lead ECG placement and acquisition](#), [EtCO₂ Monitoring](#)
- Assess LVAD for function – Refer to algorithm
- Assess patient for non-LVAD related illness

- Waveform EtCO₂

- [Vascular access](#)
- If a FLOW ALARM is noted, 20mL/kg [Isotonic Crystalloid](#) fluid challenge.

- Monitor and interpret ECG
- If FLOW ALARM still active after fluid bolus – [Norepinephrine](#) (Levophed)

Pearls:

- Normal methods of obtaining blood pressure will not work on LVAD patients; will need to use other means of determining circulatory state, such as EtCO₂. Be aware that SPO₂ will be unreliable in the presence of continuous flow devices.
- To determine LVAD function, place a stethoscope over the apex of the heart and listen for a humming sound; the absence of humming indicates device failure. First attempt to restart the device, refer to flowchart.
- When transporting an LVAD patient, ensure that extra batteries and power module (if available) are transported with the patient. If changing batteries, change only one at a time.
- Utilize the patient's family members or caretaker as they may have specific knowledge of the function of the LVAD. Keep them with the patient.
- ECGs should be routinely performed, even if the complaint does not appear cardiac in nature. ECGs will likely demonstrate mechanical interference from the device. Contact LVAD coordinator even if complaint does not seem cardiac in nature.



Cardiac: Care of the LVAD Patient

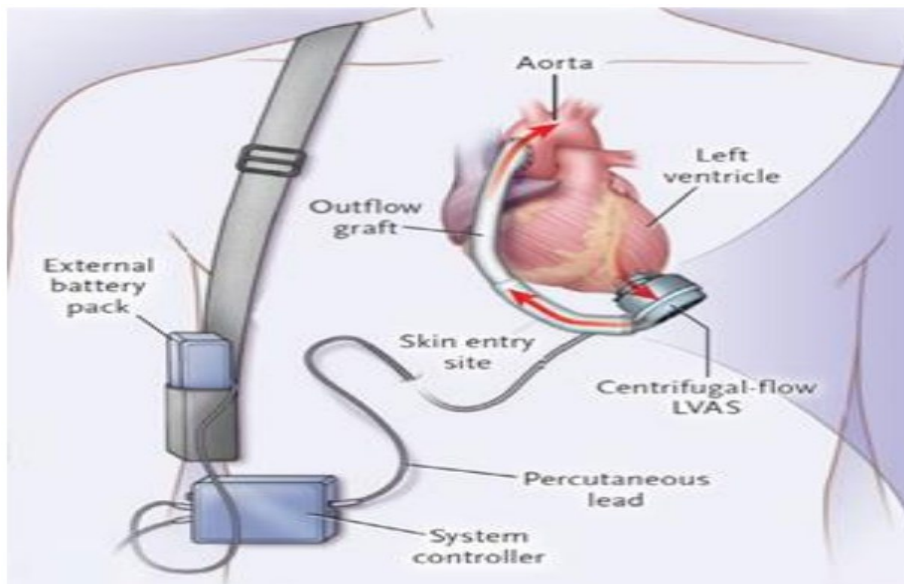
Transport patient to their LVAD facility if possible, however it may be necessary to transport the patient to the closest LVAD facility:

- St. David's Heart Hospital of Austin – LVAD coordinator emergency line – (512) 660-9564
- Ascension Seton MC – Austin – LVAD Coordinator emergency line – (512) 797-4260

LVAD related emergencies:

- **Unstable Arrhythmias:** Common complications in LVAD patients. Symptoms vary and rarely, patients may be asymptomatic with ventricular arrhythmias. Symptomatic arrhythmias should be treated according to standard COGs.
- **Unexplained Hypotension:** Usually occurs from conditions resulting in decreased pre-load such as [sepsis](#), [hypovolemia](#), [bleeding](#), and tamponade. This will likely trigger a low-flow alarm.
- **Bleeding Complications:** All LVAD patients are on blood thinners. [Epistaxis](#) is the most common bleeding complication. All bleeding should be treated with supportive interventions and efforts to cease bleeding if possible. **Do not administer TXA.**
- **Stroke:** LVAD patients are at a higher risk for stroke and intracranial hemorrhage. Maintain a high suspicion for neurologic symptoms and perform a thorough history and physical assessment.
- **Infection:**
 - **LVAD-Specific:** Involve the driveline, pocket, pump, and/or cannula
 - **LVAD-Related:** Infections not involving the LVAD itself, but can have different implications or management in a LVAD patient (infective endocarditis, bacteremia)
 - **Non-LVAD Related:** Not affected by the presence of an LVAD (UTI, pneumonia)
- **Pump thrombosis:** Suspect with lasting increases in pump power by >50% in addition to high flow estimates. These patients should be promptly transported to the closest LVAD center.
- **Pump stoppage or failure, driveline trauma:** Pump stoppage occurs when there is a complete loss of power to the pump due to loss of battery power, disconnection of power leads, or disconnection of driveline system to controller. These patients will present with acute heart failure symptoms and may need vasopressors if in cardiogenic shock.

A typical left ventricular assist device (LVAD) consists of a pump connected to the cardiac apex and ascending aorta via an inflow cannula and outflow graft, a percutaneous driveline that exits the skin on the right, and a system controller that is typically work on a belt or vest. Power to the controller and pump is provided by external batteries or a power-based unit.





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